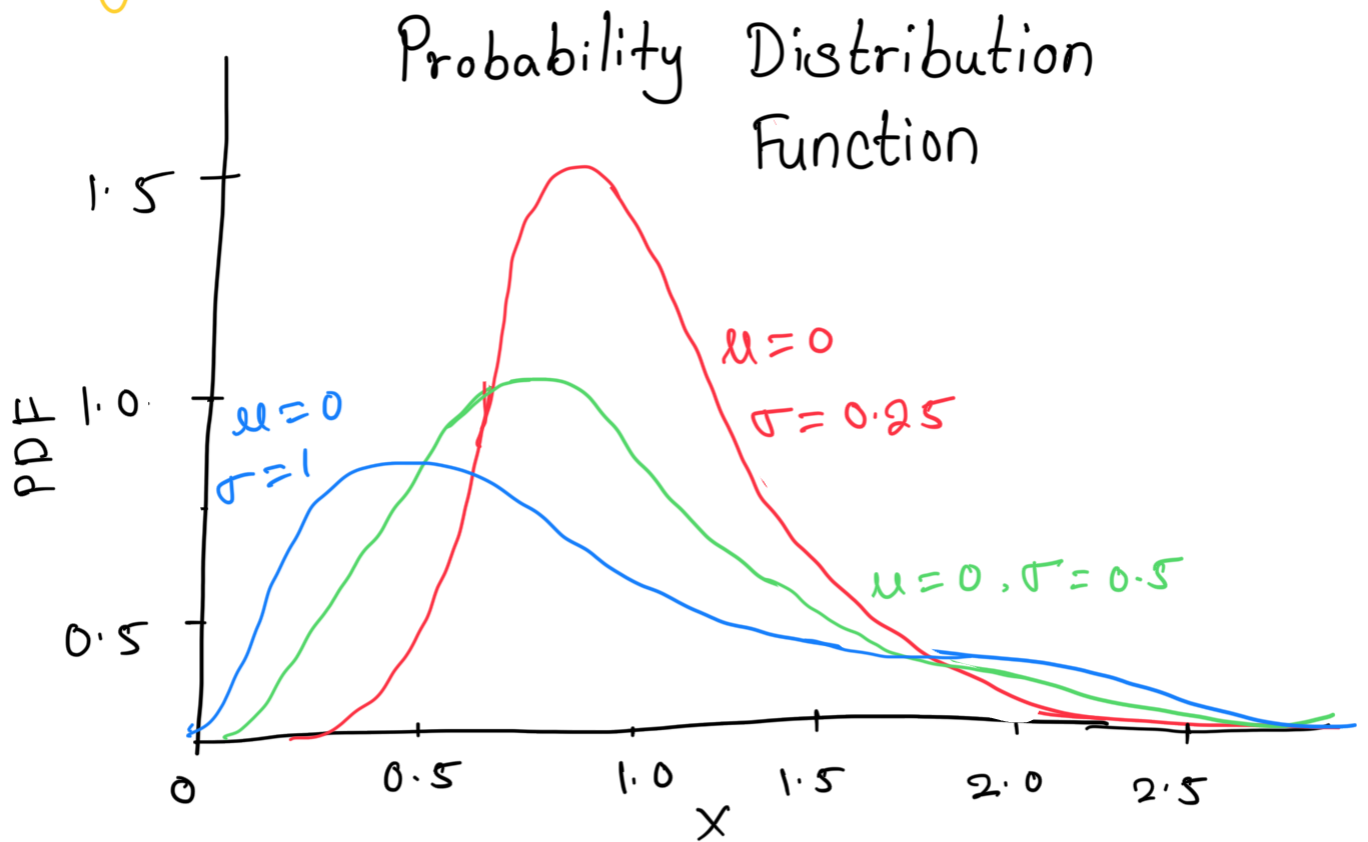


Log Normal Distribution



A random variable X follows a log-normal distribution if the $\log$ of X is normally distributed

Why does this matter
Normal distribution allows

1) Negative values

2) Symmetry around the mean

But many real-world quantities

1) Can not be negative

2) Are skewed to the right

3) Grow multiplicatively, not additively

Log-normal distribution solves this problem

shape & properties of a log-normal distribution

A log normal distribution has following characteristics :

1) Only positive values

$x > 0$ always

Zero & negative values are impossible

2) Right-skewed distribution

most observations are small to moderate

3) Mean, median, mode are different

$$\text{mode} < \text{Median} < \text{Mean}$$

4) Taking the log reduces skewness

IMP
★ ★ ★

Log transform helps only when the data is positive & multiplicative in nature.

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When positive, right-skewed variable is log-transformed, it may become normally distributed; exponentiating a normal variable produces a log-normal, right-skewed distribution

Log transformation is designed to

compress large values, so it is naturally suited for right-skewed positive data.

We use log normal because it

1) Fixes skewness

Right-skewed data violates model assumption

2) Stabilizes variance

Large values don't dominate learning

3) Improves model performance

Linear models work better in log spaces

4) Makes errors interpretable

Errors becomes percentage errors, not absolute errors.

Adding random effects gives a normal distribution; multiplying random effects gives a log-normal distribution.

so normal & log-normal both are types of probability distributions

1) Normal Distribution

- A normal distribution describes how a random variable behaves
- It is symmetric & bell shaped
- $X \sim \text{Normal}(\mu, \sigma^2)$

i.e X Follows normal distribution

2) Log-normal Distribution

- A log-normal distribution also describes how a random variable behaves

- The variable is positive & right skewed
- It is defined using normal distribution
- $X \sim \text{Log-normal}(\mu, \sigma^2) \Leftrightarrow \log(X) \sim \text{Normal}(\mu, \sigma^2)$

* Normal & log-normal are probability distributions, while log transformation is a data transformation used to change the scale & shape of data.